IN A LOCALLY CONNECTED SPACE THE OPEN SUBSETS ARE LOCALLY CONNECTED AND THE CONNECTED COMPONENTS ARE OPEN

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The definition of local connectedness through a fundamental system of connected neighbourhoods, and the notion of a connected component. Set up in the author's section on connected components, immediately before the counted statement. Fixes the hypothesis and the two conclusions of the statement.

Lemma 0.1. *Let X be a topological space. If X is locally connected, then*

(1) *any open subset of X is locally connected, and*
(2) *the connected components of X are open.*

So also the connected components of open subsets of X are open. In particular, every point has a fundamental system of open connected neighbourhoods.

Proof. For all $x \in X$ let write $\mathcal{N}(x)$ the fundamental system of connected neighbourhoods of x and let $U \subset X$ be an open subset of X. Then for all $x \in U$, U is a neighbourhood of x, so the set $\{V \in \mathcal{N}(x) | V \subset U\}$ is not empty and is a fundamental system of connected neighbourhoods of x in U. Thus U is locally connected and it proves (1).

Let $x \in \mathcal{C} \subset X$ where $\mathcal{C}$ is the connected component of x. Because X is locally connected, there exists $\mathcal{N}$ a connected neighbourhood of x. Therefore by the definition of a connected component, we have $\mathcal{N} \subset \mathcal{C}$ and then $\mathcal{C}$ is a neighbourhood of x. It implies that $\mathcal{C}$ is a neighbourhood of each of his point, in other words $\mathcal{C}$ is open and (2) is proven. □

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